

CAR SALES ANALYSIS

Bright Motors

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Dataset Bright Car Sales · 558,811 Records

Period January 2014 – July 2015

Agenda

01

Data Overview & Quality

Dataset summary, null analysis and data gaps acknowledged

02

Revenue by Make & Model

Top performing car brands and models by total revenue

03

Regional Performance

Sales volume and revenue across US states and regions

04

Vehicle Insights

Body style, transmission, mileage and condition analysis

05

Profit Margin Analysis

Margin tier distribution and pricing strategy assessment

06

Recommendations

Strategic actions for the new Head of Sales

Data Overview & Quality

558,811

Total Records

16

Columns

12

Null Columns

2014–15

Date Range

Null Analysis by Column

Column	Null Count	% Missing
transmission	65,352	11.7%
body	13,195	2.4%
condition	11,794	2.1%
trim	10,651	1.9%
model / make	~10,300	1.8%
color / interior	749 each	0.1%
mmr / sellingprice	12 each	<0.1%

Data Gaps Acknowledged

No cost_price column — MMR (Manheim Market Report) used as proxy for profit margin

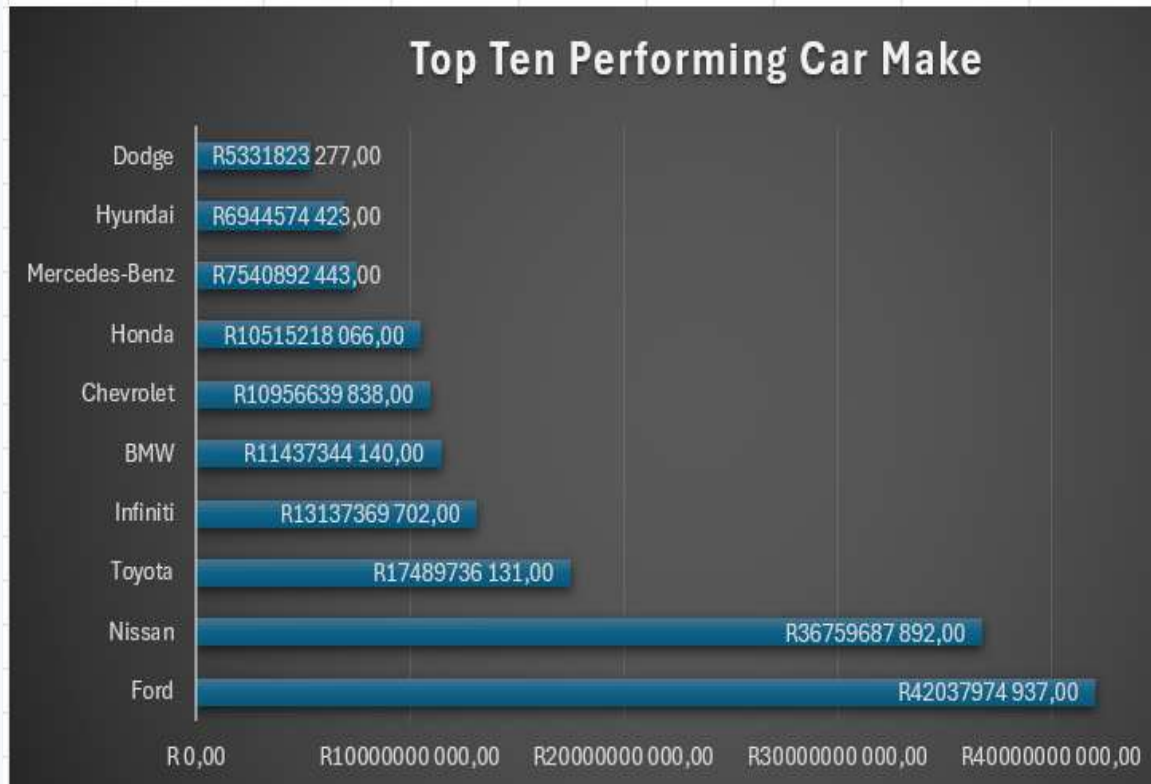
No units_sold column — derived via COUNT(*) OVER (PARTITION BY make, model, saledate)

No fuel_type column — transmission can be used as closest available proxy but was not included/dealt with.

12 records with null MMR & selling price — retained but excluded from margin calculations

65,352 unknown transmission values — unknown segment outsells manual (data quality risk)

Revenue by Car Make (Top Ten Performing)



Key Insights

Ford

Leads in both units sold and total revenue — the clear dominant brand across all regions

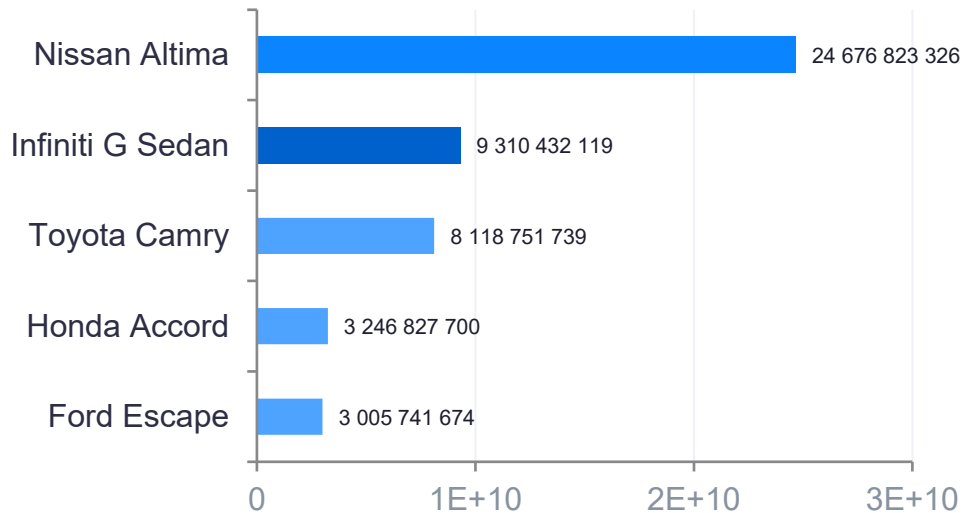
Nissan & Toyota

Strong runner-ups with combined revenue rivalling Ford, indicating balanced market demand

BMW & Mercedes

Lower volume but high average selling price — key premium segment with margin potential

Top 5 Car Models by Total Revenue



Revenue Summary

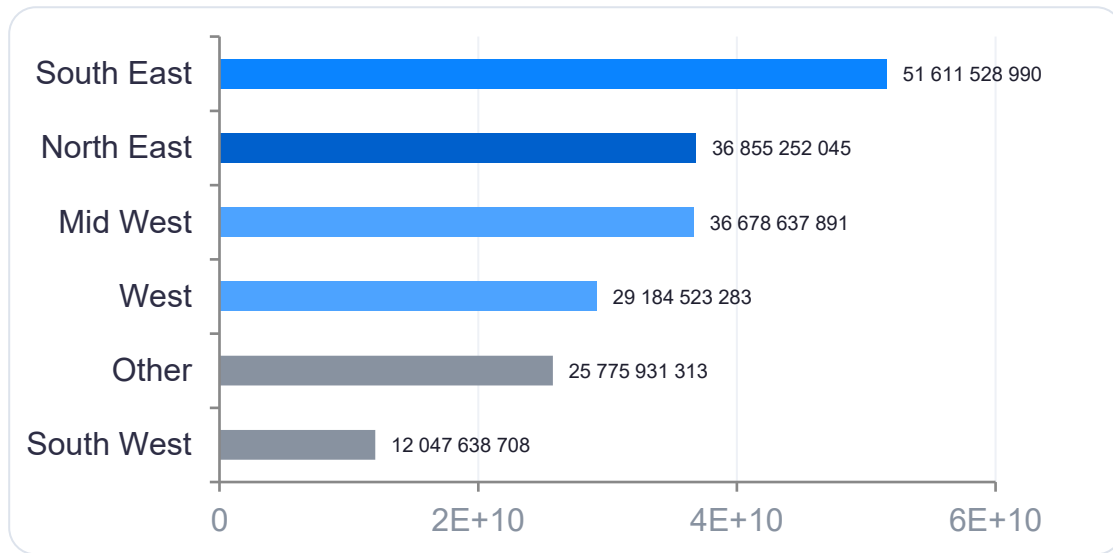
Rank	Make	Model	Total Revenue (ZAR)
1	Nissan	Altima	R 24,676,823,326
2	Infiniti	G Sedan	R 9,310,432,119
3	Toyota	Camry	R 8,118,751,739
4	Honda	Accord	R 3,246,827,700
5	Ford	Escape	R 3,005,741,674

Key Findings

The Nissan group dominates the top 5, with the Nissan Altima (R24.6B) and Infiniti G Sedan (R9.3B) collectively accounting for over 70% of the top 5 combined revenue. This reflects strong brand ecosystem performance — the Altima capturing the high-volume mainstream segment while the Infiniti G Sedan commands the premium end. The 2012 model year was the single strongest contributor across makes. Toyota Camry ranks third at R8.1B, showing consistent demand but declining revenue per unit over time. Honda Accord and Ford Escape round out the top 5 at R3.2B and R3.0B respectively — competitive in volume but significantly behind the Nissan group.

Regional Performance by Make

Total Revenue by Region (ZAR)



Key Findings

South East is the highest revenue region at R51.6B — led by Nissan (R11.2B). North East and Mid West follow closely at R36.8B and R36.6B. The West is a premium market led by Infiniti and BMW rather than volume brands. South West is the weakest region at R12B which is below other region at R25B.

Top Make per Region

South East	R11,258,252,615
Nissan	
North East	R9,542,035,729
Nissan	
Mid West	R11,848,246,970
Ford	
West	R5,554,882,495
Infiniti	
South West	R3,967,924,897
Ford	

Data Quality

Duplicate make names detected: Mercedes-Benz/mercedes, VW/vw, Land Rover/landrover — requires SQL CASE fix.

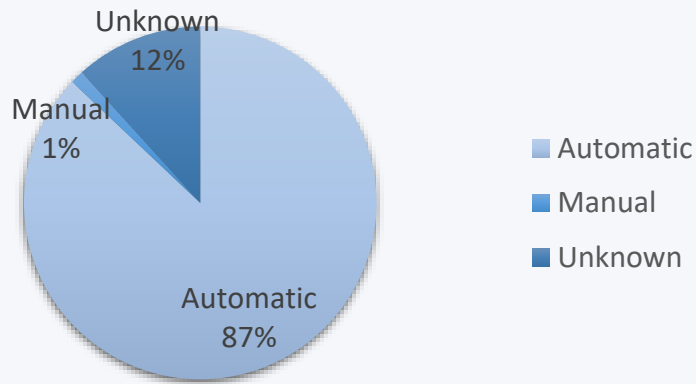
Vehicle Insights — Body Style & Transmission



Sedan dominates at 63.6% of all units sold — nearly 4x more than SUV at 15.7%. Together they account for 79.3% of total sales, confirming these are the core body styles of the business. Pickup Truck ranks third at 7.0% — consolidated from multiple cab configurations in the raw data. Minivan and Coupe are minor segments at 3.8% and 3.2% respectively.

Revenue Share by Transmission

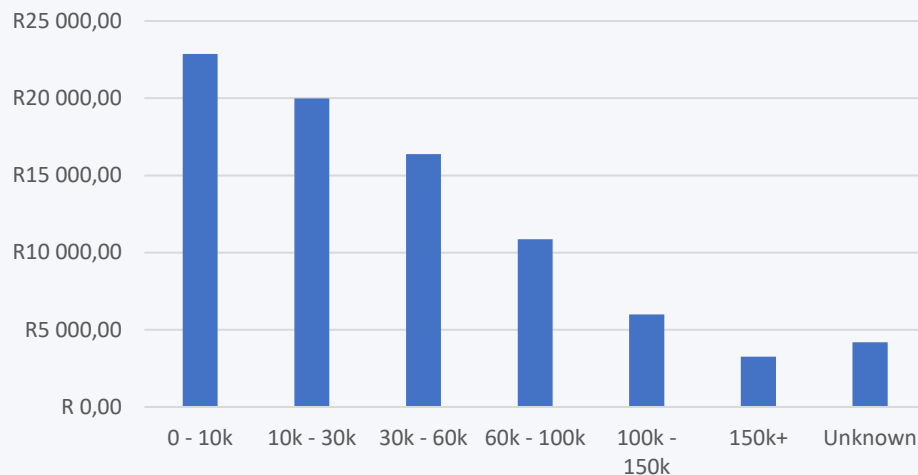
Total Revenue by Transmission



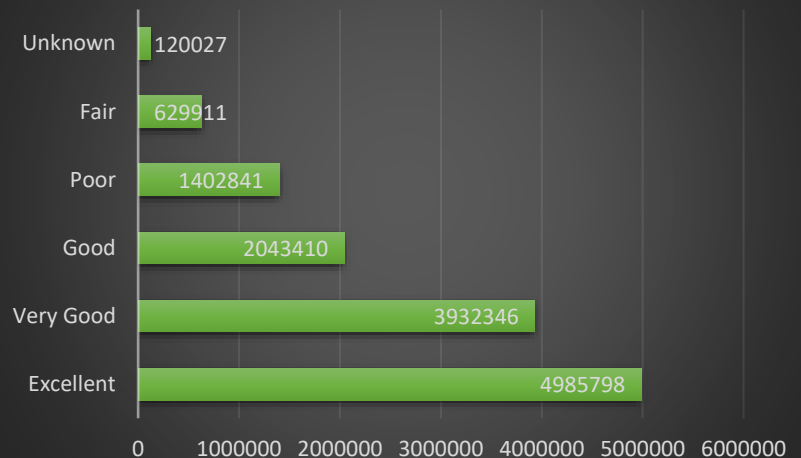
Automatic transmission dominates revenue at 87%, with Manual contributing just 1% — confirming customer preference is overwhelmingly automatic. The 12% Unknown transmission is a direct result of 65,352 missing records in the dataset.

Mileage & Condition Performance

Average Selling Price by Mileage bucket

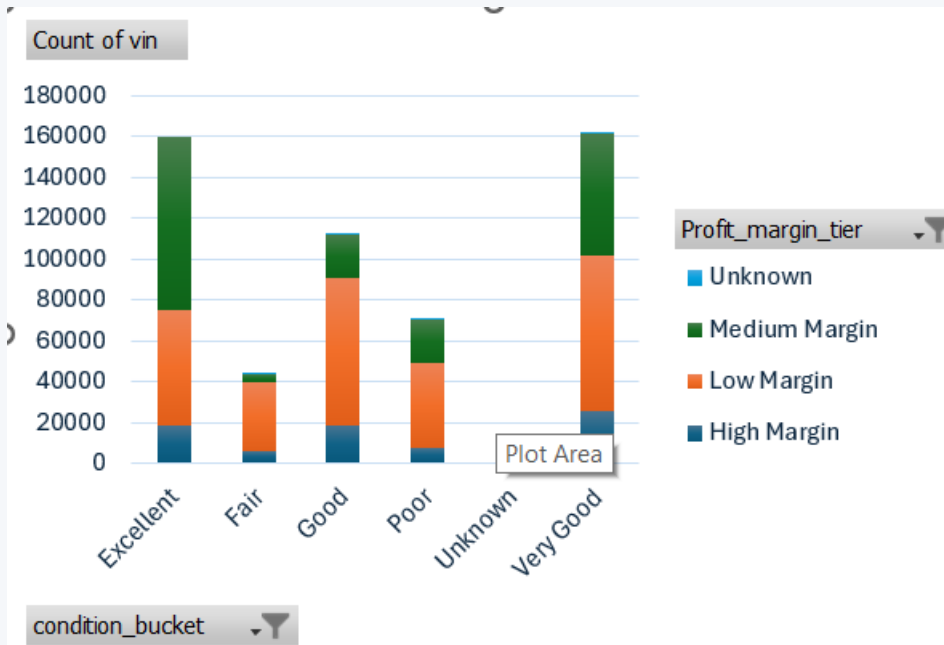


Units Sold By Vehicle Condition



- As mileage increases, average selling price drops consistently, from R22,859 for near-new vehicles under 10k miles down to R3,255 for high mileage vehicles above 150k miles. This confirms a strong inverse relationship between mileage and vehicle value. The Unknown bucket at R4,188 sits close to the 150k+ range, suggesting these may be high mileage vehicles with missing odometer data.
- Vehicles in excellent condition have the highest sales volume, showing that buyers generally prefer well-maintained vehicles. However, poor-condition vehicles sell more units than fair-condition vehicles, suggesting strong demand from budget-conscious buyers looking for lower-priced options.
- ❖ Black color leads in both unit sold and revenue, White and Silver are also major contributors to total revenue.

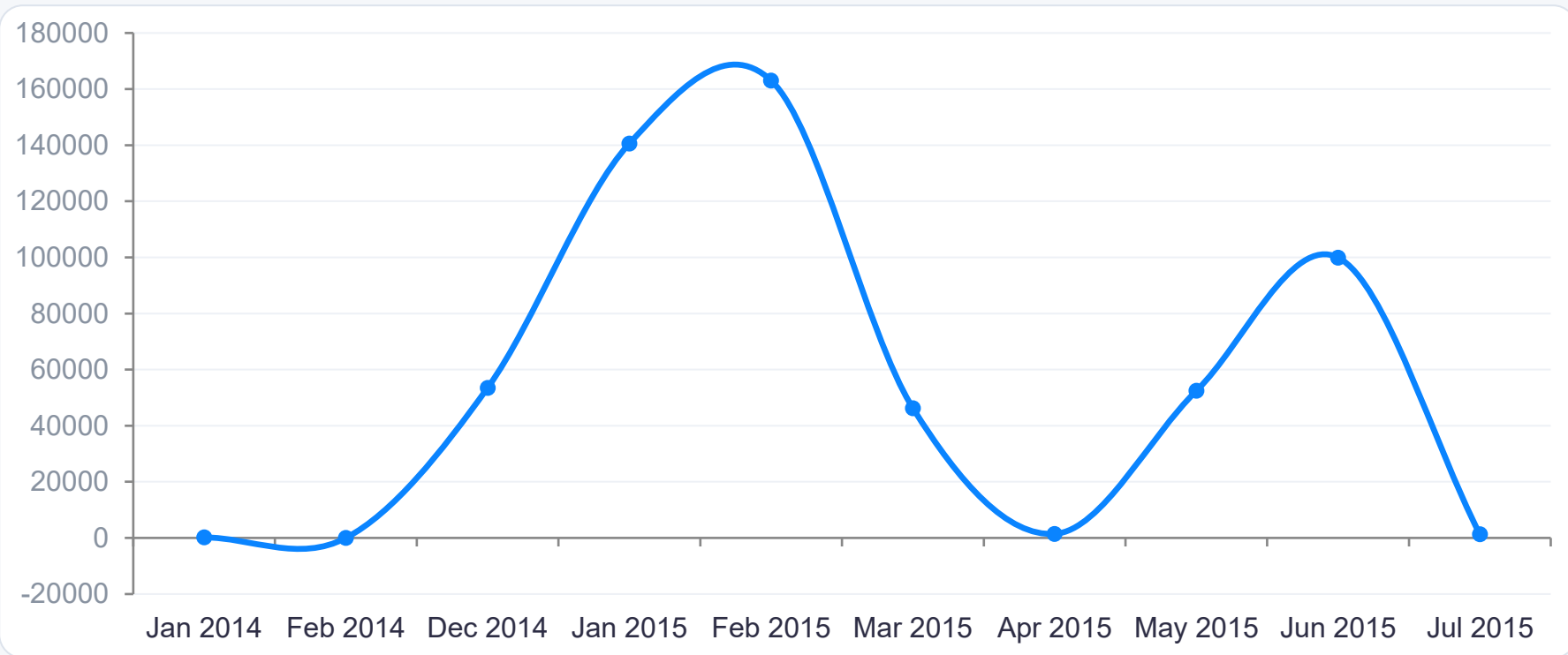
Vehicle Distribution by Condition & Profit Margin



Vehicle condition strongly influences sales volume, with Excellent and Very Good cars selling the most units. However, profit margin performance is dominated by Low Margin transactions across all condition types, suggesting that most vehicles are sold at or below market value regardless of condition. This indicates pricing pressure in the market rather than condition-driven premium pricing.

Note: MMR used as cost price proxy. 12 records with null MMR excluded from margin calculations which could be replaced by average.

Sales Trend by Period



Feb 2015 recorded peak volume. Jan & Feb 2014 show minimal activity — data collection likely began mid-period. Strong Q1 2015 suggests seasonal demand patterns worth monitoring.

Strategic Recommendations

01

Fix Pricing Strategy

All mileage segments show negative margins. Review and adjust pricing against MMR benchmarks to recover lost revenue across the entire inventory.

02

Vehicle Acquisition Strategy

Prioritize acquisition of post-2010 vehicles with under 30k miles — these command the highest selling prices and likely the best margins. Avoid acquiring pre-2005 vehicles with over 100k miles as they generate minimal revenue and likely negative margins.

03

Double Down on Ford & Nissan

Ford (R42B) and Nissan (R36.8B) are the clear national revenue leaders. Prioritise and expand inventory of these two makes across all regions.

04

Expand South East Presence

South East is the #1 revenue region at R51.6B . FL and GA represent the highest ROI opportunity for dealership expansion.

05

Leverage West as Premium Market

The West is led by Infiniti (R5.6B) and BMW (R5.2B) — not volume brands. Target premium inventory and marketing strategies specifically for West Coast dealerships.

06

Prioritize Automatic Transmission

Prioritize automatic transmission vehicles in inventory. Mandate transmission capture at point of sale to eliminate the 12% unknown gap and get a true picture of the split.

Thank You

Questions & Discussion
